TECTONICS, VOL. 3, NO. 2, PAGES 179-200, APRIL 1984

PALEOMAGNETIC ASSESSMENT OF OROFLEXURAL DEFORMATION IN WEST-CENTRAL NEVADA AND SIGNIFICANCE FOR EMPLACEMENT OF ALLOCHTHONOUS ASSEMBLAGES

John W. Geissman and James T. Callian

Department of Geology, Colorado School of Mines

John S. Oldow

Department of Geology, Rice University

Stephan E. Humphries

Department of Geology, Colorado School of Mines

Abstract. Abrupt westerly deflections of north-south trending facies boundaries, the isopleth of $(^{87}Sr/^{86}Sr)^{o} = 0.076$, and arcuate structural trends of Paleozoic, Mesozoic, and Cenozoic age in the west-central Basin and Range Province, have been explained by (1) post-Early Jurassic oroflexural folding and shear, (2) the existence of a pre-Late Jurassic resistant block along the western continental margin, possibly representing late Precambrian breakup, and (3) regional flexure ("Mina deflection") prior to intrusion of Late Cretaceous plutons which crosscut flexure-related structural trends. Paleomagnetic data gathered from Late Cretaceous plutons and remagnetized metasedimentary rocks along and north of the northern margin of the zone of east-west structural trends are not in support of significant oroflexural folding (and attending major clockwise rotation) since Late Cretaceous time. The data (Water Canyon pluton: D = 1°, I = 57°, α_{95} = 4°, k = 9, N = 132; Gunmetal pluton: D = 354, I = 60, α_{95} = 6, k = 21, N = 29; East Garfield Hills pluton: D = 349, I = 74, α_{95} = 11, k = 14, N = 12; La Panta pluton: D = 22, I = 66, α_{95} = 10, k = 16, N = 13; Gillis Canyon pluton: D = 354, I = 50, α_{95} = 4, k = 20, N = 50; and folded units of the Excelsior Mountains: D = 309, I = 79, α_{95} = 7, k = 12, N = 37) are dispersed from expected Cretaceous directions for the Mina deflection region, but they do not conform to a model of major systematic regional clockwise rotation. The generally small deflections of unit means from expected directions could be explained by irregular components of Late Cenozoic tilting and/or rotation. The paleomagnetic data do not deny the possibility of pre-Late Cretaceous oroclinal bending. Nevertheless, regional structural analysis complimented by the paleomagnetic results indicate that the Mina deflection is in all likelihood a primary feature whose geometry reflects the late Precambrian fragmentation of western North America. Structures related to the accretion of allochthonous assemblages between Middle Devonian and Cretaceous time in this part of the Cordillera also reflect the primary nature of this feature.

Copyright 1984
by the American Geophysical Union

Paper number 4T0009.
0278-7407/84/004T-0009$10.00

INTRODUCTION

In west-central Nevada, Late Cenozoic extensional structures are superimposed upon a complex assemblage of deformed pre-Tertiary rocks. Understandings of the Mesozoic paleogeography of this region, the kinematics of regional folding and thrusting, the ages of deformation, and regional structural/stratigraphic correlations are all critical to deciphering the tectonic history of the western United States. Much work has addressed these subjects [Albers, 1967; Ferguson and

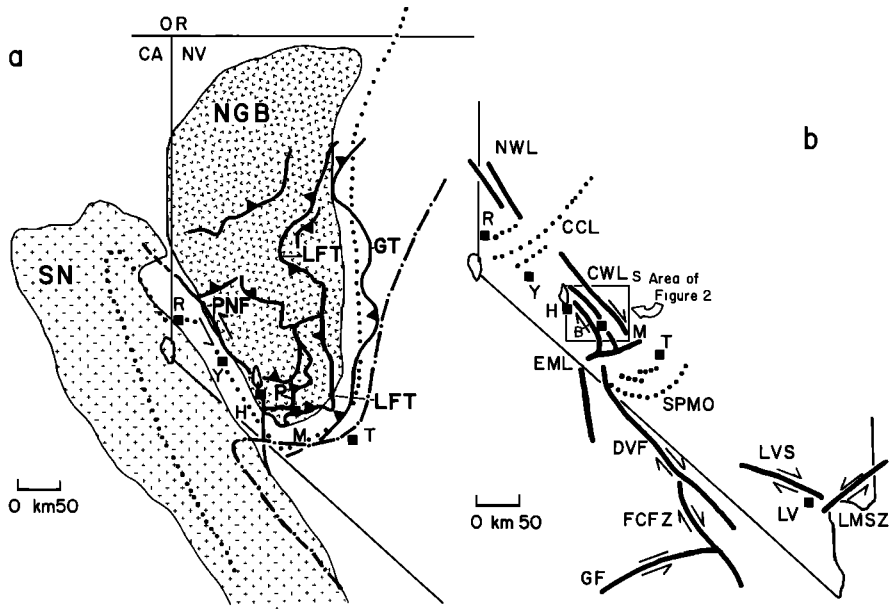


Fig. 1. Maps showing generalized patterns of major structural elements and/or physiographic features of Mesozoic (Figure 1a) and Cenozoic (Figure 1b) age and the distribution of pre-Cenozoic basement rocks (Figure 1a) in the western Basin and Range Province. (a) Mesozoic features: dotted line, 0.706 $^{87}Sr/^{86}Sr$ isopleth; dotted/dashed line, eastern limit of Permo-Triassic volcanic rocks; solid lines, thrust or strike-slip faults (GT, Golconda thrust; LFT, Luning-Fencemaker thrust; P, Pamlico thrust; PNF, Pine Nut sinsitral fault); SN, Sierra Nevada; NGB, Mesozoic marine province of the northwest Great Basin. R, Reno; Y, Yerington; H, Hawthorne; M, Mina; T, Tonopah. (b) Cenozoic features: solid lines, strike-slip faults (NWL, northern Walker Lane; CWL, central Walker Lane; EML, Excelsior Mountains lineament; DVF, Death Valley fault zone; FCFZ, Furnace Creek fault zone; GF, Garlock Fault; LVS, Las Vegas shear zone; LMSZ, Lake Mead shear zone); dotted lines, physiographic features (CCL, Carson City lineament; SPMO, Silver Peak-Palmetto-Montezuma oroflex). LV, Las Vegas; S, Shoshone Mountains sampling site of Kluger-Cohen [1981]; b, New Boston/Blue Ribbon sampling site of N. B. Harris and M. O. McWilliams (manuscript in preparation, 1984). Modified from Hardyman [1978], Albers [1967], Kistler and Peterman [1973], Kistler et al. [1981], Oldow [1983b], and Seidenstricker [1983].

Muller, 1949; Muller and Ferguson, 1939; Oldow, 1978a, b, 1981a, b, 1982, 1983a, b, 1984a, b; Oldow et al., this issue; Reilly et al., 1980; Speed, 1978a, b, 1979, 1983; Speed and Kistler, 1980; Wetterauer, 1974, 1977]. Near Mina, Nevada (Figure 1a), trends in the distribution of and structures in pre-Tertiary rocks and the Sr 0.076 isopleth, thought to locate the approximate position of the western margin of Precambrian North America [Kistler, 1983; Kistler et al., 1981; Farmer and DePaolo, 1983], are deflected abruptly westward. Also, well-defined are abrupt changes in trends of Cenozoic normal and strike-slip faults and earthquake epicenters (Figure 1b). The origin of this curvilinear feature, termed the Mina deflection by Wetterauer [1974, 1977], has been debated for a long time [Muller and Ferguson, 1939; Ferguson and Muller, 1949; Albers, 1967: Oldow, 1983b, 1984b]. The basic argument concerns whether the Mina deflection represents a pre-Late Jurassic embayment in the western margin of North America [Muller and Ferguson, 1939; Ferguson and Muller, 1949; Oldow, 1983b, 1984b] or the product of significant right-lateral oroflexural folding [Albers, 1967; Wetterauer, 1974, 1977]. The

19449194, 1984, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC003i002p00179 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

timing of oroflexural deformation, if it did occur, is unclear as well. Albers [1967] suggested that the folding postdated widespread intrusion of Cretaceous plutons in the Mina area. Wetterauer [1974, 1977], on the other hand, proposed that the deformation predated Cretaceous plutonism, because the plutons crosscut folds presumably related to oroflexure. Certainly, other models for the Mina deflection, involving, for example, prolonged dextral shear along several northwest trending structures, could be invoked. Clarification of the origin and age of the Mina deflection is of obvious importance in interpreting the kinematics of Mesozoic tectonism and establishing Mesozoic paleogeography in the region.

Numerous Late Cretaceous siliceous plutons in the Mina area offer a potential test of the hypotheses bearing on the Mina deflection. We have studied the paleomagnetism of several plutons, as well as some deformed (and remagnetized) layered rocks, in hopes of showing that the magnetizations characteristic of the plutons are essentially indistinguishable from expected directions for the stable craton or systematically rotated, in a clockwise fashion, away from expected directions. Any systematic rotation of directions could presumably be ascribed to oroflexural deformation postdating the plutons. We emphasize that the present analysis investigates the possibility of oroflexural deformation only in post-Late Cretaceous time. Earlier components of deformation cannot be evaluated with magnetization data from Cretaceous plutons. The effectiveness of this study hinges on whether local structural elements related to Basin and Range extension (e.g., simple tilting of crustal blocks) can be differentiated from rotational components that might be associated with the formation of the Mina deflection. We will address this question in detail in the discussion section.

Our observations largely support a primary, nonoroflexure/rotation origin for the geometry of the Mina deflection. They have clear and significant ramifications for Paleozoic and Mesozoic tectonism and paleogeography in west-central Nevada. We have no doubt that local, shear-related deformation may have occurred in at least portions of the Basin and Range. Gillett and Van Alstine [1982a, b] have documented local rotation of crustal blocks forming the Desert Range and have related the deformation to Tertiary dextral slip along the Las Vegas shear zone. Data supporting the lack of regional clockwise rotation of the Mina deflection area, which is mandated by oroflexural deformation, further imply that orientations of structures related to contraction of the region in Paleozoic (e.g., Roberts Mountains thrusting) and Mesozoic (e.g., Luning-Fencemaker thrusting) time [Oldow et al., this issue] do not require correction for regional rotations of potentially large magnitudes.

GENERAL GEOLOGY

The general geology of west-central Nevada consists of a basement complex of latest Paleozoic to Cretaceous allochthonous metasedimentary and metavolcanic rocks intruded by granitic plutons. The plutons range from earliest Jurassic to latest Cretaceous age. The basement units are unconformably overlain by sequences of middle Oligocene to Miocene ash-flow tuffs, Miocene andesite flows and dikes, and Miocene-Pliocene basalt flows. Data were collected from plutons in three localities, the first near the town of Mina, the second southeast of Hawthorne, and the third northeast of Hawthorne, and from folded metasedimentary rocks in the Excelsior Mountains southwest of Mina (Figure 2).

The first locality includes the Excelsior Mountains, Garfield Hills, and the Pilot Mountains, where the basement rocks have been studied extensively [Ferguson and Muller, 1949; Muller and Ferguson, 1939; Oldow, 1978a, b, 1981a, b, 1983a, b, 1984b; Reilly et al., 1980; Seidensticker, 1983; Speed and Kistler, 1980; Wetterauer, 1974, 1977]. Layered rocks of this area are predominantly of shallow marine origin that were deposited in and around a "successor" basin (NGB in Figure 1a) formed in response to the cessation of earlier contraction along the Roberts Mountains and Golconda thrust systems and bounded on the west by a volcanic arc terrane and on the east by continental uplands [Oldow, 1981a; Speed, 1978b]. Many of these layered units are allochthonous; their present distribution reflects major Mesozoic thrusting and Cenozoic deformation. In the Mina area (Figure 3), a major east-west trending thrust belt involves layered rocks of various ages and compositions [Oldow, 1981a, b, 1983a, b]. The Luning thrust

19449194, 1984, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC003i002p00179 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

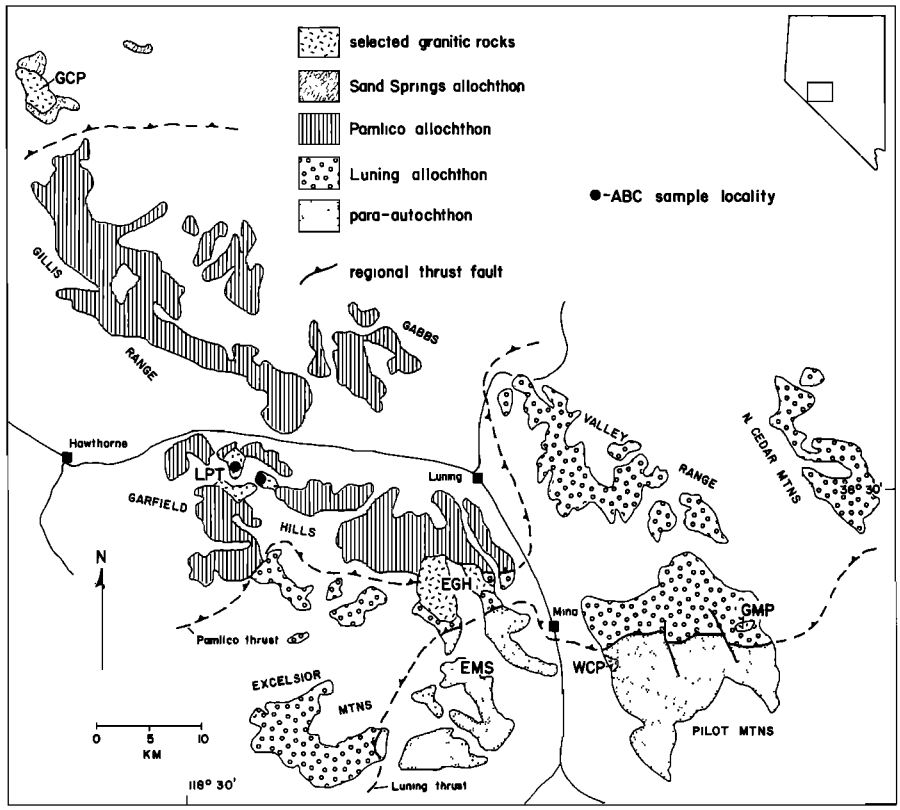


Fig. 2. Generalized distribution of pre-Tertiary layered rocks in the Mina-Hawthorne region, along the northern margin of the Mina deflection, west-central Nevada. All sampling localities are located and are as follows: WCP, Water Canyon pluton; GMP, Gunmetal pluton; EGH, East Garfield Hills pluton; LPT, La Panta pluton; GCP, Gillis Canyon pluton; EMS, Excelsior Mountains syncline. Sampling sites in the La Panta pluton locality are shown.

separates units of the Luning and Pamlico allochthons to the north from autochthonous and/or parautochthonous units to the south. The Luning autochthon is composed of the Permian Mina formation, which is overlain by a complex of Mesozoic terrigenous-clastic and volcanic rocks which constitute two assemblages [Speed, 1977, 1978a; Speed and Kistler, 1980]. Folded siliceous volcanic rocks of the younger assemblage have yielded a Rb-Sr isochron age of 103 ± 5.7 m.y., while the older assemblage has yielded a Rb-Sr isochron age of 142 ± 17 m.y. [Speed and Kistler, 1980]. These units are involved in deformation associated with the emplacement of the Luning thrust and all structures are crosscut by plutons ranging in age from 100 to 70 m.y.B.P.

The Gillis Range (the third locality) consists largely of intensely faulted and usually tilted Cenozoic volcanic rocks, but on the western flank of the range, allochthonous Triassic metasedimentary and metavolcanic rocks, equivalent in age to the Luning Formation, are exposed [Hardyman, 1980]. One major pluton crosscuts these units and is well exposed in Gillis Canyon. It is believed to be of Cretaceous age [Hardyman, 1980].

Plutons of dioritic to granitic composition occur throughout western Nevada [Carlson et al., 1975; Speed and Kistler, 1980; John, 1983]. Speed and Kistler [1980] suggested that west-central Nevada was on the eastern margin of the Late Cretaceous Sierra Nevada batholith. The plutons sampled in the Mina area (Figures 2 and 3) are largely granite to quartz monzonite in composition. Complex contact

19449194, 1984, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC003i002p00179 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

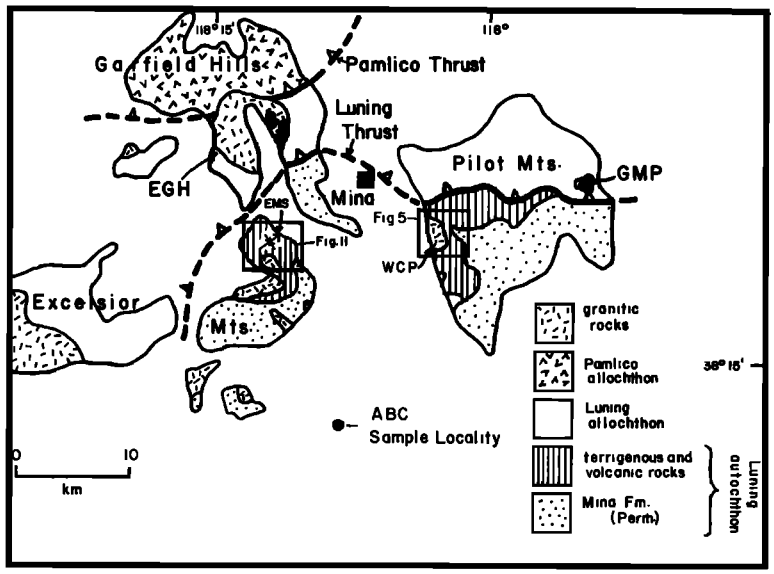


Fig. 3. Generalized geologic map of the Mina area, after Speed and Kistler [1980]. Sampling sites in the East Garfield Hills (EGH) and Gunmetal (GMP) pluton localities are also shown. The locations of the Water Canyon pluton (WCP) and the Excelsior Mountains syncline (EMS) localities are also given.

relations suggest that at least some of these plutons intrude their own volcanic cover (e.g., Speed [1980] and our own observations). We summarize available geochronologic age data for the sampled units in Table 1.

Structural elements of the west-central Basin and Range Province, as well as the above stratigraphic relations, are of importance in assessing the nature and/or timing of the Mina deflection. Cenozoic structures are dominantly high-angle faults; in much of the Basin and Range these define a north-northeast grain. The study area lies within and close to the so-called Walker Lane, a physiographic subprovince of the Basin and Range characterized as a northwest trending zone of equant-shaped ranges. Throughgoing structures in the Walker Lane consist of northwest striking, high-angle faults which locally exhibit large components of right-lateral slip [Hardyman, 1978; Molinari, 1983]. In the vicinity of Mina, the northwest trending faults either are replaced by east-northeast trending sinistral faults or curve westerly to east-northeast orientations.

Along the western margin of the Walker Lane, high-angle normal faults and, at least locally, listric normal faults generally exhibit northerly strikes [Hardyman, 1978, 1980]. The change in orientation of strike-slip faults (Figure 1) corresponds with a curvilinear belt of microseismic activity [Ryall and Priestly, 1975]. Also, focal mechanisms in this area are characterized by the absence of normal-slip events at depths greater than 10 km [Vetter and Ryall, 1983], unlike most areas in the western Basin and Range Province. These features appear to be manifestations of the Mina deflection. Albers [1967] associated the Mina deflection with many Cenozoic structures and features and thus inferred that most of the oroclinal bending followed Cretaceous plutonism.

Mesozoic structures are complex [Oldow, 1981a, b; Oldow et al., this issue; Wetterauer, 1977]. Northeast trending fold axes and superimposed northwest trending fold axes are abundant in basement rocks in the area [Oldow et al., this issue], and both fold sets are associated with thrusting. Wetterauer [1977] suggested that northwest trending fold axes correspond with the northwest trending bisector of the limbs (or straight segments) of the Mina deflection (Figure 1a) and were thus formed during oroflexural bending. Because Cretaceous plutons

19449194, 1984, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC003i002p00179 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1. Geochronologic Data For Sampled Units, Mina Deflection Region

Locality	Age, m.y.	Method	Reference
Water Canyon pluton, Pilot Mountains	69	K-Ar, biotite	Schilling [1965]
	<142 ± 17	Rb-Sr, whole rock	Speed and Kistler [1980]
Gunmetal pluton, Pilot Mountains	83.4 ± 2.1	K-Ar, biotite	D. Grabher (personal communication, 1983), Union Carbide
East Garfield Hills pluton, southeast Garfield Hills	94.5 ± 2.5	K-Ar, biotite	Everndon and Kistler [1970]
La Panta pluton, Garfield Hills	92.1 ± 3.4	K-Ar, biotite	this study and John [1983]
Gillis Canyon pluton, northwest Gilles Range	Late Cretaceous (?)		Hardyman [1980], John [1983]
Excelsior Mountains syncline (clastics), northeast Excelsior Mountains	>95.8 ± 2.9 (Silver Dyke pluton)	K-Ar, biotite	Garside and Silberman [1978]
	<103 ± 5.7	Rb-Sr, whole rock	Speed and Kistler [1980]

crosscut these folds, Wetterauer concluded that formation of the feature preceeded plutonism.

Recent investigations addressing the tectonic origin of the Mina deflection at least support the original hypothesis of Muller and Ferguson [1939] for an irregularly shaped western margin of continental North America embayed on the Mina-Hawthorne area. Oldow [1982, 1984a] has shown that structures in the upper plate of the Roberts Mountains allochthon within and to the south of the Mina deflection area indicate similar east-to-west thrust transport directions as those in north-central Nevada. According to Oldow [1983b, 1984a], the allochthon was not rotated on the western limb of the Mina deflection, as predicted by the Albers and Wetterauer models. In addition, a really restricted development of cross-folds in the Roberts Mountains allochthon near the Mina deflection suggests that an east-west trending crustal irregularity predated the Antler orogeny. Structural data thus clearly suggest a nontectonic origin for the bend. The paleomagnetic data presented here offer further, compelling evidence that the Mina deflection, whatever its origin, was formed prior to Late Cretaceous plutonism.

METHODOLOGY

We report paleomagnetic data from 35 sites in five plutons and a tightly folded succession of parautochthonous sedimentary rocks. At each site in the six sampling localities, generally seven or more independently oriented samples were collected by standard means. All orientations were made by using a magnetic compass; all NRM intensities recorded were sufficiently low (less than 1.0 A.m) to preclude local field disturbances and associated orientation problems.

Measurements of the remanent magnetization were done with a Schonstedt SSM-1a spinner magnetometer interfaced with a Cromemco system two microprocessor and a Hewlett-Packard (HEDS-500) 500 count/rotation optical shaft encoder. Progressive

19449194, 1984, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC003i002p00179 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

alternating field (AF) demagnetization utilized a Schonstedt GSD-1 single-axis demagnetizer; thermal demagnetization was done in a Schonstedt TSD-1 furnace in which the specimen temperature was monitored by using a thermocouple probe set inside a "synthetic" specimen during each run.

The results of each demagnetization run were plotted on orthogonal demagnetization diagrams. Magnetization components that were removed could be analyzed with standard vector subtraction techniques, as linear decay of magnetization components usually occurred. As a number of petrologically and texturally different units have been investigated, magnetization components characteristic of individual units or sampling areas were not always removed over one particular range of demagnetizing peak fields or temperatures. Thus optimum demagnetizing peak fields or temperatures were not established for a particular unit.

Isothermal remanent magnetization (IRM) acquisition, backfield demagnetization, and Curie temperature (Js-T) tests were performed to aid in identifying the magnetic mineralogy of the units investigated.

Most of the units investigated have been dated or, at least in the case of the folded metasedimentary rocks, have the age of the magnetization well constrained (Table 1). We report here for the first time an age determination for the La Panta stock in the Garfield Hills (Table 1). Samples were collected from our paleomagnetic sampling sites and biotite separates were made by one of us (J. S. O.). Kreuger Geochron Laboratory performed the K-Ar analysis in 1977. John [1983] included these data on his summary map.

A brief description of each sampling region is given, together with the respective paleomagnetic data.

PALEOMAGNETIC DATA

Water Canyon Pluton

The Water Canyon pluton intrudes parautochthonous terrigenous and volcanic rocks of the Luning autochthon in the west central Pilot Mountains [Nielson, 1964; Oldow, 1978a, 1981a; Speed and Kistler, 1980]. Its northern margin is a few hundred meters south of the southern boundary of the Luning thrust (Figures 3 and 4). Samples were collected from 19 sites (Figure 4), 12 representing different phases of the pluton and 7 located in surrounding preplutonic rocks of the "Water Canyon" and "Telephone Canyon" assemblages [Oldow, 1981a; Speed and Kistler, 1980]. Schilling [1965] reported a K-Ar biotite age of 69 m.y. for the pluton. Deformed rhyolitic crystal tuffs that were sampled directly south of the pluton yield a whole rock Rb-Sr age of 142 ± 17 m.y. [Speed and Kistler, 1980].

Demagnetization of samples from all lithologies is generally straightforward, with a component of northerly declination and moderate to steep positive inclination (or antipode) over a wide range of peak fields or blocking temperatures (Figures 5a-5d). The paleomagnetic data have been divided into two groups: (1) coarse-grained intrusive units of the Water Canyon pluton and (2) surrounding units, largely metavolcanic and metasedimentary rocks (Figures 6a and 6b, respectively). The directions of major components removed during demagnetization are scattered and we attribute this to two possible effects. First, the data shown in Figure 6a imply that the Water Canyon pluton cooled through at least one field reversal and therefore that the dispersion of data from the intrusive units, especially from samples with characteristic magnetizations of relatively discrete blocking temperatures, may reflect anomalous directions related to field transitions. Second, our sampling of surrounding units includes a number of lithologies (rhyolitic breccia, ash-flow tuff, volcanic conglomerate and sandstone, chert-volcanic breccia, dikes, etc.) of differing structural orientation. We assume that these units were in large part remagnetized during intrusion of the Water Canyon pluton, and the data in Figure 6b imply that remagnetization spanned at least one field reversal. Our assumption may not be completely correct, however, as terrigenous volcanic rocks in Telephone Canyon (near sampling sites 12 and 13, Figure 4) do not show Rb-Sr isotopic reequilibration as a result of intrusion of the Water Canyon pluton [Speed and Kistler, 1980]. Structural corrections may thus need to be applied to at least some of the data, but the magnitude, sense, and timing of deformations are all difficult to estimate. At present, we treat all data as if the rocks were in situ to establish an overall direction for this sampling region.

19449194, 1984, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC003i002p00179 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

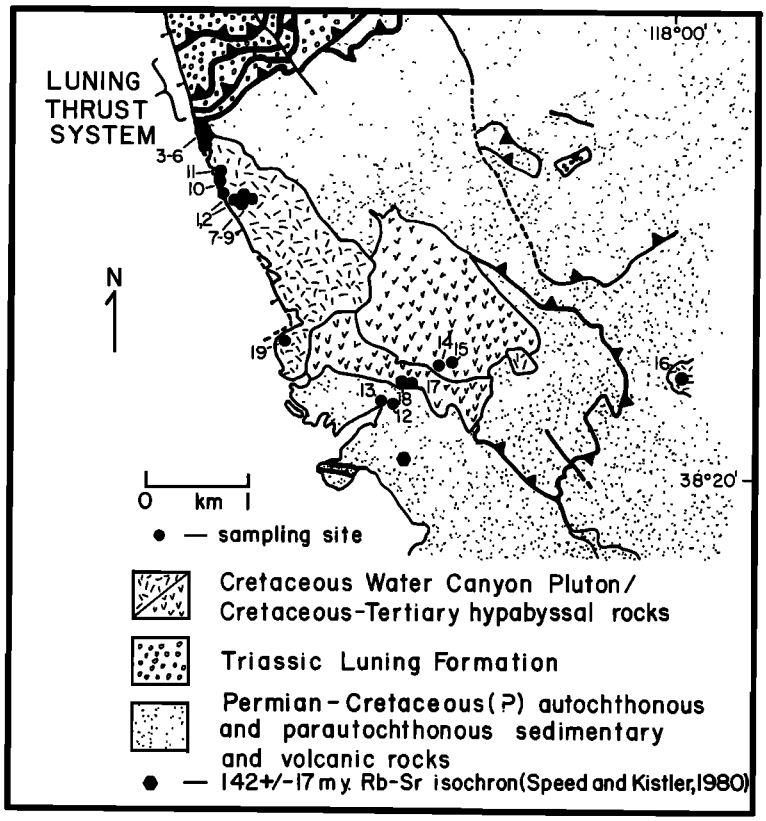


Fig. 4. Generalized geologic map of the Water Canyon pluton and surrounding units, western Pilot Mountains. After Speed [1980].

Gunmetal Pluton

A stock of quartz monzonite intrudes carbonate units of the Luning allochthon in the east-central Pilot Mountains [Nielson, 1964; Oldow, 1978a]. Its K-Ar date of 83.4 ± 2.1 m.y. (D. Grabher, personal communication, Union Carbide, 1983) suggests a Late Cretaceous age. Samples were collected at two sites in the pluton itself and at one site in metasomatized calc-silicate and marble units at the contact (Figure 3). All samples from the pluton contain a magnetization of northerly declination and steep to moderate positive inclination, which is removed over moderate peak fields and distributed temperatures (Figures 5e, 5f). Samples of the contact units contain similar components of magnetization, but their magnetization properties (higher coercivities and blocking temperatures) are different from those of the Gunmetal pluton samples. All magnetization data are plotted in Figure 6c. The directions obtained from the contact units are statistically indistinguishable from data from the intrusive, implying that the host units have been at least locally remagnetized. The normal polarity of the characteristic magnetization is consistent with emplacement of the pluton during the Cretaceous Long Normal interval.

East Garfield Hills Pluton

This pluton intrudes upper plate rocks of both the Luning and Pamlico allochthons (Figure 3). The pluton, dated as 94.5 2.5 m.y. [Everndon and Kistler, 1970], crosscuts the Pamlico thrust, establishing an upper bound on the age of deformation. The pluton intrudes the Permian Black Dyke formation, the Triassic Luning formation, and the Jurassic-Cretaceous Dunlap formation. Samples were collected from five sites in the pluton (Figure 3). A component of magnetization with a northerly declination and a moderate positive inclination is characteristic of this pluton (Figure 7a) and is removed over a wide range of peak fields and temperatures in

19449194, 1984, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC003i002p00179 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

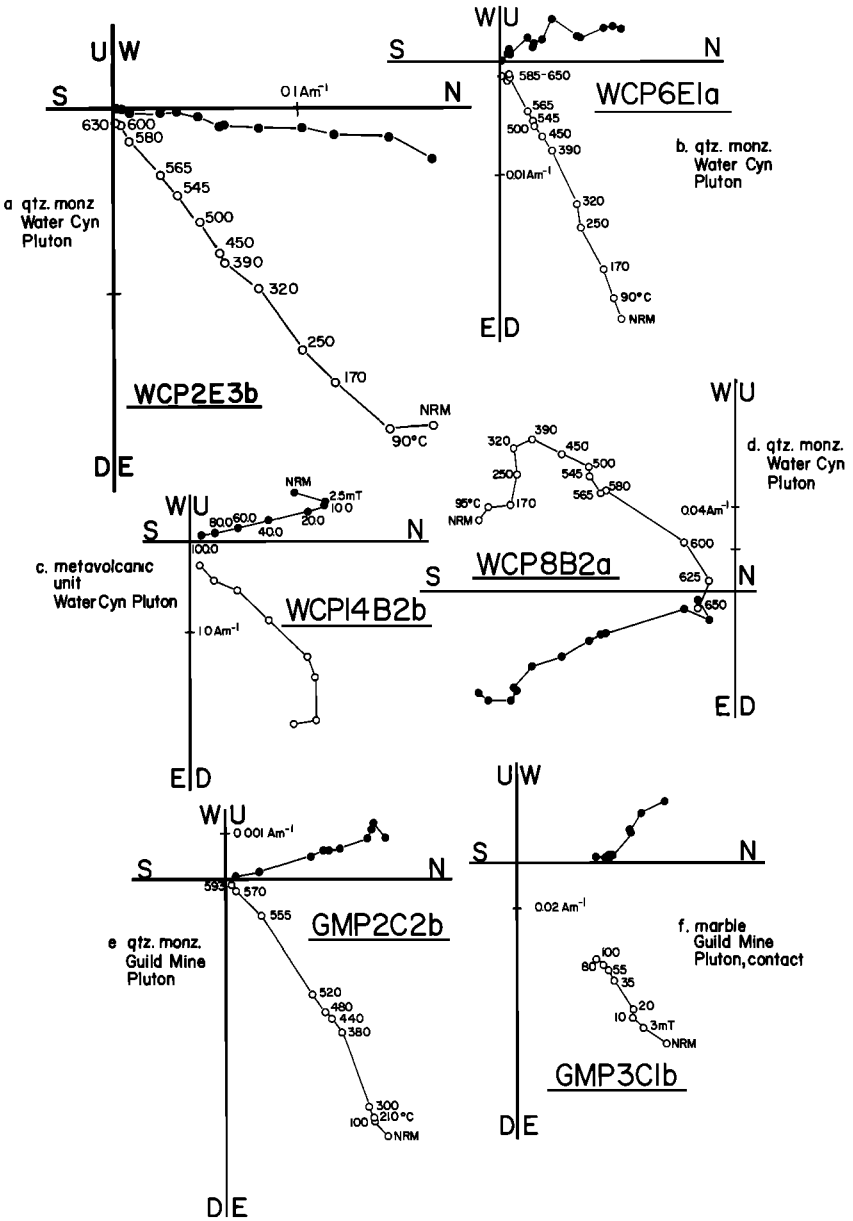


Fig. 5. Orthogonal progressive demagnetization diagrams [Zijderveld, 1967] of the end point of the magnetization vector during alternating field or thermal demagnetization of specimens of Mina deflection area plutons, metasedimentary rocks, and metavolcanic rocks. In these and all other progressive demagnetization diagrams, the data are always plotted with respect to in situ (geographic) coordinates. Open (solid) symbols represent projections of the end point of the magnetization vector onto the vertical (horizontal) plane.

demagnetization. The normal polarity of the magnetization (Figure 6d) is consistent with the pluton being emplaced during the Cretaceous Long Normal interval.

La Panta Pluton

The La Panta Pluton intrudes the Triassic Pamlico formation and undifferen-

19449194, 1984, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC003i002p00179 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

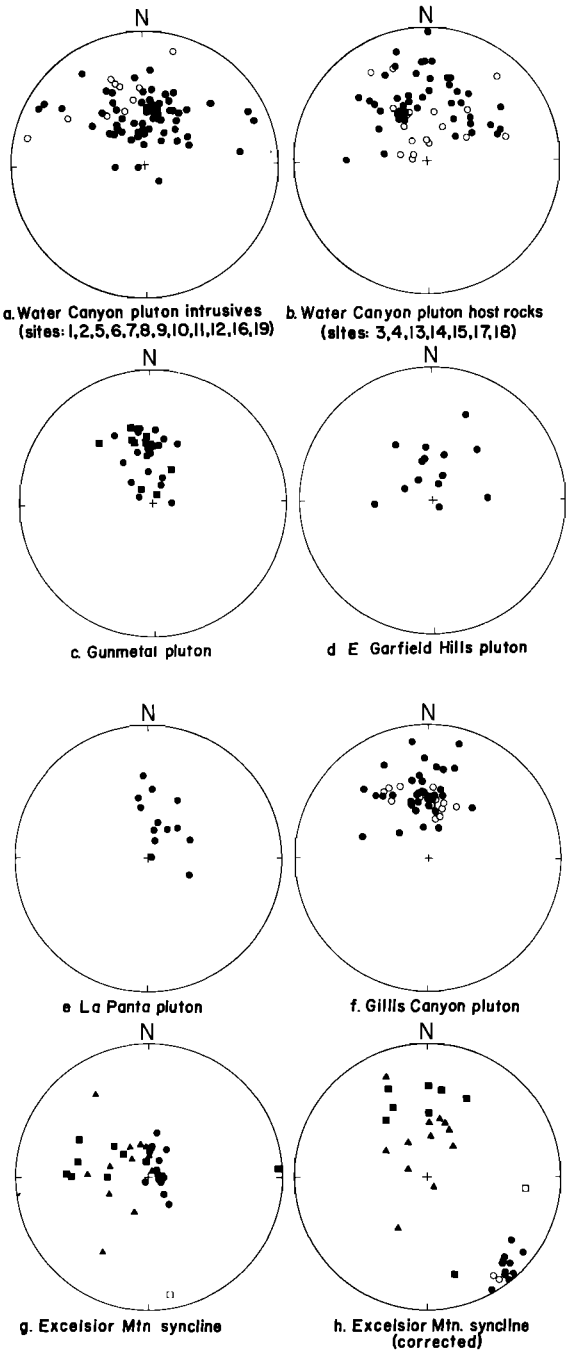


Fig. 6. Paleomagnetic data from each of the Mina deflection sampling localities. In each diagram, sample components of magnetization isolated during progressive demagnetization are plotted on equal area stereographic projections. The components plotted are characteristic of each sampling locality. Note that components with southerly declination and negative inclination have been inverted through the center to show the clustering of all data about a normal polarity direction; their originally negative inclination is still denoted by an open symbol. Solid (open) symbols refer to lower (upper) hemisphere projections. In all diagrams except Figure 6h, the data are plotted in in situ coordinates.

tiated Triassic and Jurassic Sunrise-Gabbs formations in the upper plate of the Pamlico allochthon [Oldow, 1978b, 1981a] (Figure 2). The pluton is relatively poorly exposed and often deeply weathered; only two sites have been sampled to date. Our previously unreported ages of 92.1 ± 3.4 and 97.8 ± 3.6 m.y. suggest that the pluton is of mid-Cretaceous age.

Demagnetization of most samples isolated a component of magnetization with a northerly declination and moderate to steep positive inclination over a wide range of peak fields or temperatures (Figure 7b). The directions of these components are plotted in Figure 6e. Their normal polarity is again consistent with emplacement during the Cretaceous Long Normal interval.

Gillis Canyon Pluton

This pluton is undated, but intrudes Triassic metasedimentary rocks equivalent in age to the Luning formation (Figure 8). The Triassic rocks probably belong to the Mesozoic Sand Springs allochthon [Oldow, 1984b]. The Gillis Canyon pluton is centrally located within the Walker Lane shear zone, and structures in overlying and surrounding Cenozoic volcanic rocks reflect a complex interplay of strike-slip and extensional faulting [Hardyman, 1978, 1980; Hardyman and Oldow, 1984]. Seven sites were sampled in the Gillis Canyon pluton (Figure 8); one site (GCP 24) included samples of metamorphosed Triassic rocks.

The Gillis Canyon pluton contains a complicated magnetization record. The last component removed, requiring moderate to high peak fields or temperatures, is either of northward declination and moderate positive inclination (Figure 7d) or southward declination and moderate negative inclination (Figure 7c). Four of the sites have either constant normal or reversed polarity, but at three sites both polarities are present. Cooling of the pluton must have spanned at least one

19449194, 1984, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC003i002p00179 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

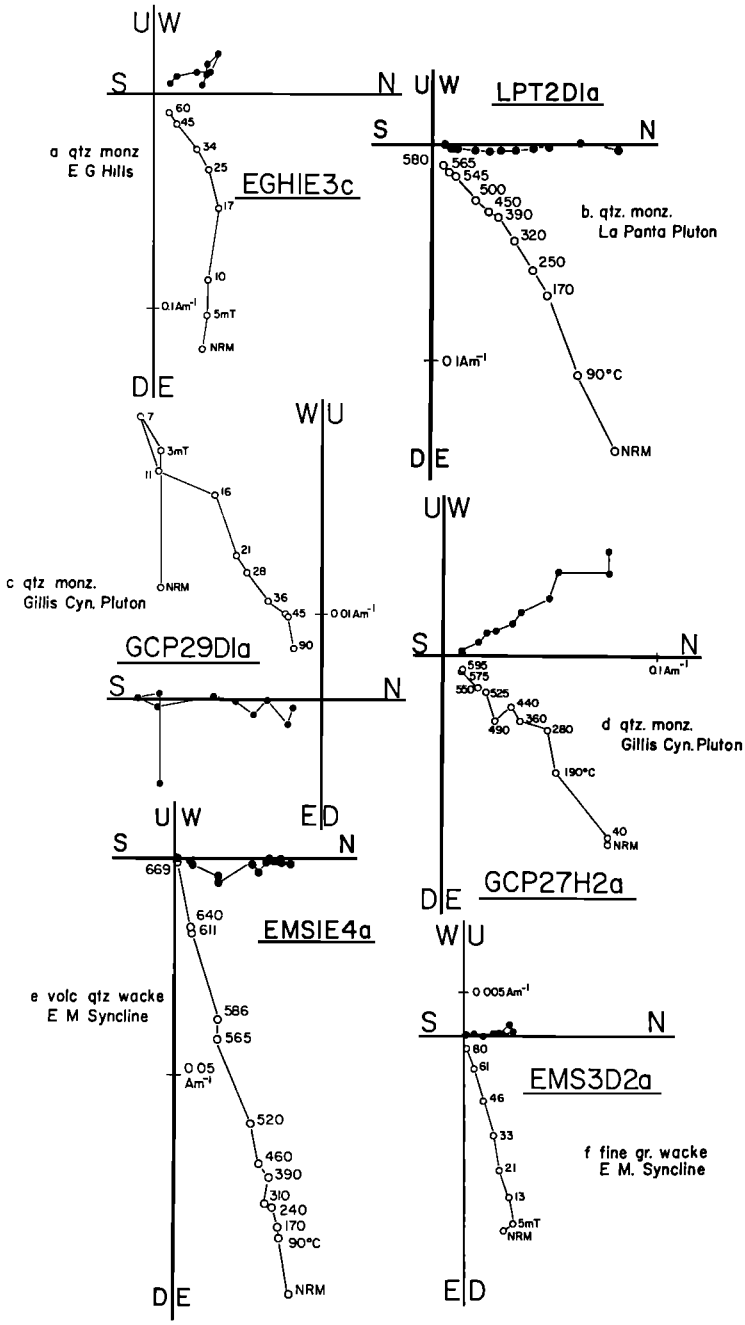


Fig. 7. Orthogonal progressive demagnetization diagrams of the end point of the magnetization vector during alternating field or thermal demagnetization of specimens of Mina deflection area plutons and metasedimentary rocks. Conventions are as in Figure 5.

polarity transition. This may imply that the pluton is younger than about 85 m.y., the age of the end of the Cretaceous Long Normal zone, but short reversals in the Cretaceous Long Normal interval have been detected. Data from the Gillis Canyon pluton are shown in Figure 6f. Reverse polarity directions are statistically indistinguishable from those of northerly declination and positive inclination.

19449194, 1984, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC003i002p00179 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

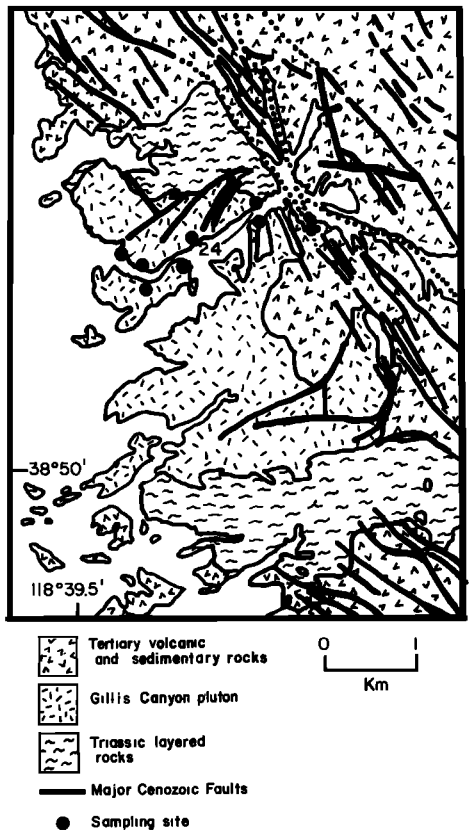


Fig. 8. Generalized geologic map of the Gillis Canyon pluton and surrounding units. After Hardyman [1980].

Excelsior Mountains Syncline

In the northeastern Excelsior Mountains, autochthonous (or parautochthonous) terrigenous sedimentary and volcanic rocks are folded into a northeast plunging syncline (Figure 9). Details of the geology of the area are given by Speed and Kistler [1980]. Samples from three sites on the northwest limb, southeast limb, and trough of the fold were collected in clastic units overlying volcanic tuffs and breccias that yielded a Rb-Sr date of 103 ± 5.7 m.y. (Table 1). Distinctive beds are lacking in the upper clastic sequence [Speed and Kistler, 1980], and thus we were unable to sample a single lithologic unit continuously around the fold.

The rocks sampled include mainly dark, fine-grained to pebbly chert-quartz volcanic wackes and these exhibited viriable, but usually univectorial, behavior in demagnetization (Figures 7e and 7f). This single component of magnetization is usually of steep positive inclination (in situ sense) with blocking temperatures reaching up to 680°C. A few samples contained principal magnetization components of anomalously shallow inclination.

In situ magnetization directions are plotted in Figure 6g, and structurally corrected directions are plotted in Figure 6h. The in situ directions are predominantly of very steep, positive inclination. No antipodal directions (those with steep negative inclinations) were recorded. Structural correction of the data from the folded units may be carried out in several ways. We assume that the plunge of the fold was acquired after folding because there is no evidence of regional tilting prior to Cretaceous folding and because the strain analysis work of Oldow [1981a] has shown that fold axes were generally formed in a horizontal orientation. The tilt component of deformation was thus removed before the limbs were restored to the horizontal. The order of the exercise is actually insignificant; the value of k for the in situ data is 12.4, while k = 2.0 for the structurally corrected data set. An F ratio test establishes that the fold test for the Excelsior Mountains fold is negative at the 99.9% confidence level. Remagnetization must have occurred after deposition and deformation of these units. We suggest that intrusion of the Silver Dyke pluton (95.8 ± 2.9 m.y., Table 1), less than 1 km south of the sampling area (Figure 9), was responsible for the remagnetization of these units. A constant normal polarity for the remagnetized metasedimentary units is consistent with emplacement of the Silver Dyke pluton during the Cretaceous Long Normal interval. We have now sampled this pluton extensively; initial paleomagnetic data from it are characterized by very steep positive inclinations and generally north to northeast declinations.

MAGNETIC MINERALOGY

Plutons

Overall, the magnetic mineralogy of the sampled plutons and their behavior during demagnetization are typical of coarse-grained granitic/quartz monzonitic intrusives, as discussed by many workers [e.g., Geissman et al., 1980, 1982b; Gromme et al., 1976; Kanter and McWilliams,

19449194, 1984, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC003i002p00179 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

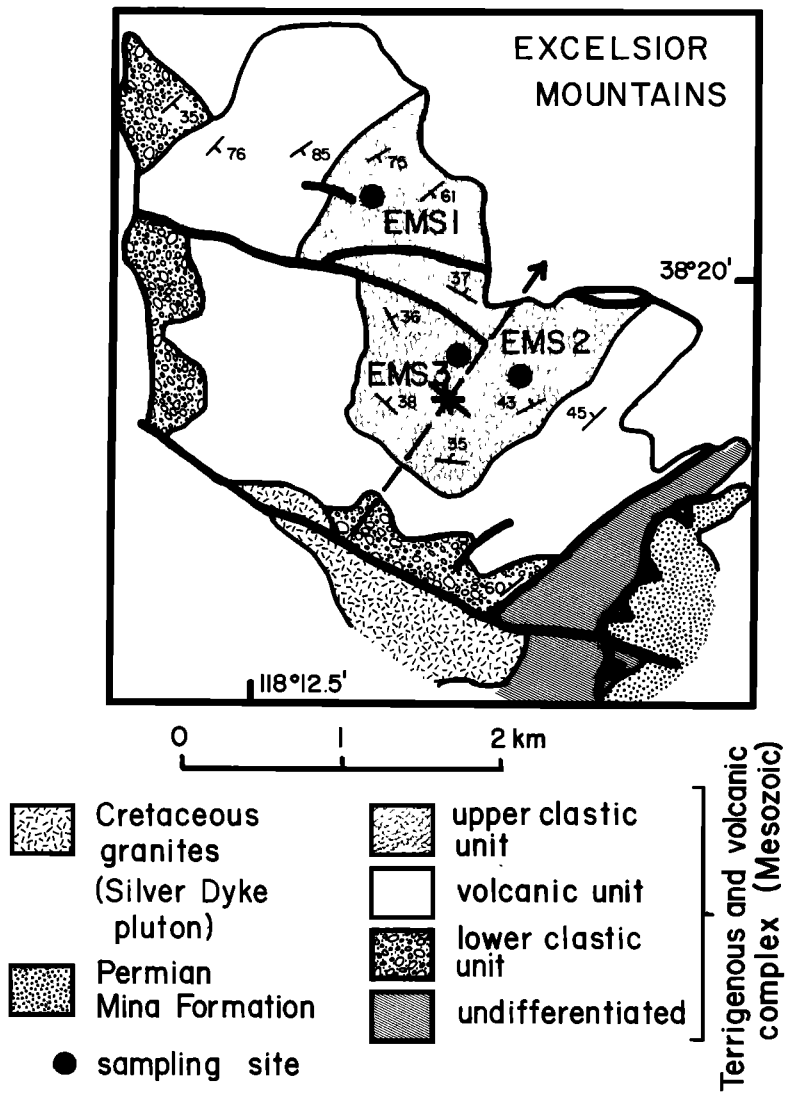


Fig. 9. Generalized geologic map of the northeast Excelsior Mountains showing the northeast plunging syncline in terrigenous sedimentary and volcanic rocks. After Speed and Kistler [1980].

1982; Puffer, 1973; Sanderson, 1974; Wu et al., 1974]. Behavior during demagnetization, showing wide ranges in coercivity and blocking temperature spectra, implies that magnetite, the major carrier of the NRM in these units as determined by Curie temperature, IRM acquisition and backfield demagnetization, and petrographic examination, is quite varied in grain size. Furthermore, our petrographic observations suggest that there is no reason to invoke a secondary origin for magnetite in these units.

Excelsior Mountains Syncline and Contact Units

Demagnetization results suggest that the remanence in the folded clastic units of the northeast Excelsior Mountains resides in both hematite and magnetite. IRM acquisition and backfield demagnetization test support these conclusions. Some proportion of the hematite present may in fact be of primary or early diagenetic origin, but this is difficult to determine by petrographic observations. We suggest

19449194, 1984, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC003i002p00179 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

that much of the magnetic mineraolgy in these units formed during and after folding in a prolonged thermochemical event. Importantly, the components of magnetization in these units clearly represent a post folding remagnetization event, probably during the Cretaceous Long Normal interval.

Demagnetization behavior and IRM acquisition tests indicate that deformed calc-silicate and marble units adjacent to the Gunmetal pluton in the eastern Pilot Mountains contain a remanence residing in both magnetite (probably of single- or pseudo-single-domain state) and hematite. Again, the components of magnetization appear to have been acquired during intrusion of the pluton.

DISCUSSION

The paleomagnetic data presented have direct bearing of the nature and timing of the formation of the Mina deflection. We raise several points to support the theory that the west stepping bend is probably a primary crustal feature. The sample localities, in plutons and deformed metasedimentary and metavolcanic rocks located along and north of the northern margin of the zone of east-west structural trends, are in opportune position to analyze the nature of the Mina deflection. Furthermore, the sample localities are in widely differing settings with respect to Late Cenozoic structural features, and, as we will discuss, it may be expected that any slight variability in magnetization directions between localities may reflect local structural complications.

While the paleomagnetic data from each locality are well grouped close to a time-averaged axial dipole field direction, several aspects of the data and the units sampled suggest that the characteristic magnetizations indeed reflect intrusive events in this area in Late Cretaceous time. The data may be used to infer amounts of Late Cretaceous and younger deformation in the area. First, each pluton and the associated magnetic mineralogy is at most only slightly altered. The only change in the primary, magmatic magnetic mineralogy of each pluton is a very weak martitization (hematite replacement of magnetite). Second, we have recorded magnetization components of both normal and reverse polarity which we believe to be of TRM origin in the Water Canyon and Gillis Canyon plutons. The very recording of field reversals by these plutons unfortunately may have added to the overall dispersion of the TRM components. Geochronologic or inferred ages for these plutons imply intrusion after the Cretaceous Long Normal interval. The East Garfield Hills, Gunmetal, and La Panta plutons and units in the Excelsior Mountains syncline display only normal polarity characteristic magnetizations, at least in agreement with earlier intrusion or remagnetization during the Cretaceous Long Normal interval. Third, secondary components of magnetization are often superimposed on the characteristic components. Fourth, storage tests in the present-day field show that the NRM in these units is not viscous in nature. Finally, the limited data we have from units in immediate contact with some plutons give positive contact tests. This is strong evidence for a primary origin for the characteristic magnetization of these plutons.

The paleomagnetic data for each sample locality, consisting of a number of individual sampling sites, are summarized in Figure 10a and Table 2. We have determined mean directions of characteristic magnetization for each locality by averaging sample directions (unit weight), not site directions. As is common for plutons or remagnetized units, our observations indicate that the dispersion of data from an individual site is as high or higher than the overall dispersion of the sample data from a particular locality sampled. We show this in Table 2 for the Gunmetal pluton. The locality mean direction determined on the basis of three site means has a very low α_{95} value in comparison to those of the site means and the mean determined on the basis of individual sample directions. This relationship holds for all other localities, and at present we prefer to determine statistics on the basis of individual sample directions.

The locality results in Figure 10a are compared with expected directions extrapolated to western Nevada coordinates for Cretaceous and earliest Tertiary time in Figure 10b. The sample locality mean directions are not coincident with expected directions for their respective time periods. To assess these discrepancies, we have calculated rotation (R) and flattening (F) values and associated errors (delR, delF) for each locality data set following Beck [1980] and Demarest [1983] (Table 3). The assumed age of the magnetization for each locality has been round-

19449194, 1984, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC003i002p00179 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

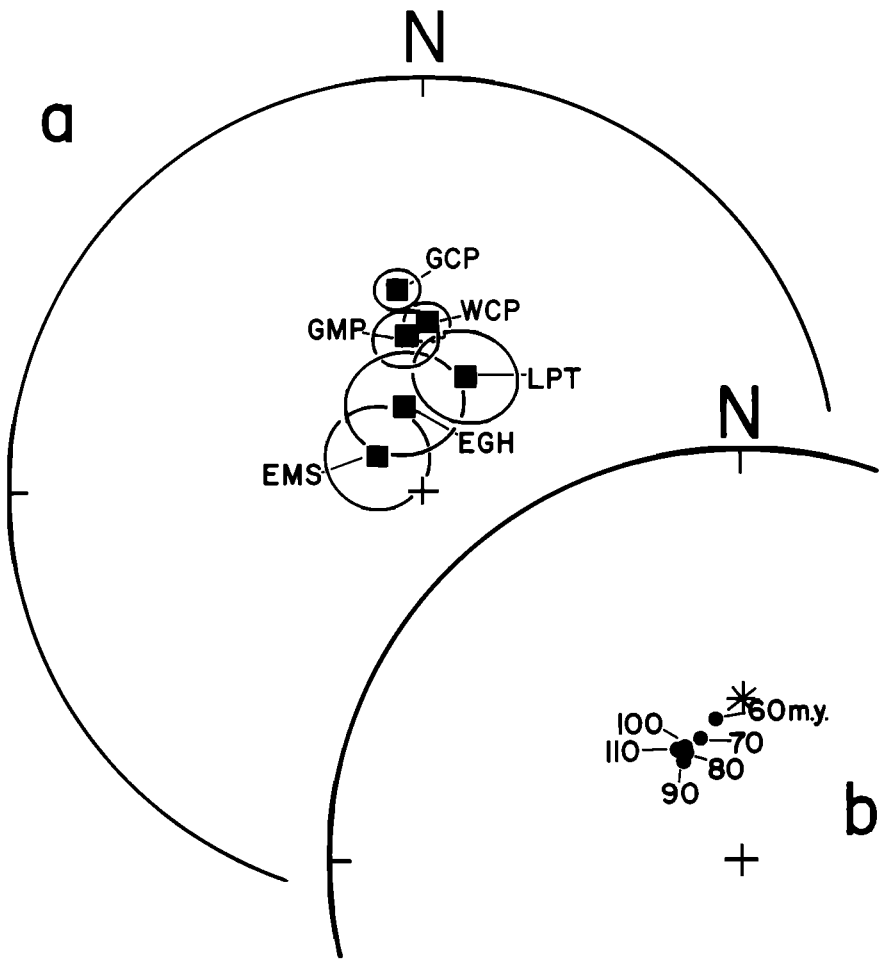


Fig. 10. Summary of the paleomagnetic data from basement plutons and remagnetized layered rocks, Mina deflection area. (a) Mean directions of sample localities (unit weight to sample directions) are given in squares and have associated projected cones of 95% confidence. (b) Expected field directions, extrapolated for west-central Nevada from paleomagnetic pole positions from Irving and Irving [1982], are given for Late Cretaceous to earliest Tertiary time. Statistics for expected directions of concern are given in Table 3. The time-averaged axial geocentric dipole field direction is given by an asterisk.

ed to the nearest 10^7 years; that of the Gillis Canyon pluton is estimated to be 70 m.y. Discordance values for this pluton would change if our estimate is incorrect. With this analysis, we make no assumptions regarding the allochthonous, or "suspect," nature of the layered units into which the plutons were emplaced or which have been remagnetized. We assume that by Late Cretaceous time the Mina region was attached to North America. In light of local-scale structural complexities related to Late Cenozoic extension in the region, our data would not necessarily support or refute this assumption.

The geographic distribution of the sampling localities with respect to east-west orientations of structure/facies trends requires that, had oroflexural deformation affected the region after plutonism and remagnetization, clockwise rotations of the sampling localities of the order of 90° would have occurred. Five out of the six sampling localities do exhibit positive rotation values (Table 3), but rotations of the required magnitude clearly are not suggested by the paleomagnetic data. In addition, flattening values of both signs are statistically significant for four of the localities.

19449194, 1984, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC003i002p00179 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2. Paleomagnetic Data, Mina Deflection Region

Locality	Number of Sites	N/N_o	Mean Declination, deg	Mean Inclination, deg	k	α_{95}, deg
Water Canyon pluton	19	132/135	1.3	57.2	9	4
Gunmetal pluton	3	29/29	354.2	59.6	21	6
site 1	...	11/11	350.2	61.6	18	10
site 2	...	9/9	359.5	57.1	30	8
site 3	...	9/9	347.8	58.6	17	11
mean (sites)	3	3/3	352.1	58.7	432	4
East Garfield Hills pluton	5	12/12	349.0	73.7	14	11
La Panta pluton	2	13/14	21.9	66.1	16	10
Gillis Canyon pluton	7	50/52	354.3	49.9	20	4
Excelsior Mountains syncline	3	37/40	308.8	79.1	12	7

N/N_o is the ratio of accepted sample directions to the total number of samples demagnetized, except for the mean for the Gunmetal pluton using three site means. The declination and inclination are given in in situ coordinates. The best estimate of the precision parameter is given by k [Fisher, 1953]. The semiangle of the cone of 95% confidence about the mean direction is given by α_{95}. Earlier data sets may be found in the works of Callian and Geissman [1982] and Oldow and Geissman [1982].

We feel that a more realistic interpretation of the paleomagnetic data from basement units of the Mina area, or any other portion of the Basin and Range Province, must logically take into account effects of local-scale late Cenozoic deformation related to extension. As above, pervasive oroflexural rotation in the Mina region should appear as systematic clockwise deflections of magnetization data, and, in fact, there is a systematic, yet statistically minor, clockwise deflection of the data from expected directions. Apparent clockwise deflections, however, can also be recording westside downtilting of crustal blocks, as long as there are no statistically significant changes in resulting inclinations. In addition, clockwise rotation of data in an episode of major clockwise rotation of a region could be removed by pervasive eastside downtilting of crustal blocks. We thus believe that Late Cretaceous magnetizations have been only slightly deflected from expected directions due to minor local-scale tilting and/or rotation of crustal blocks and that regional clockwise rotations of sufficient magnitude to produce the observed structure/facies trends have not occurred since Late Cretaceous time. Several points that bear on this hypothesis may be made.

First, late Cenozoic structural effects (e.g., local block tilting) in the Mina area are generally mild, and it is doubtful whether effects of regional clockwise rotation could have been removed by later local structural deformation. Normal faulting related to extension did not cause significant tilting of crustal blocks [Hardyman, 1978; Stewart, 1980], as in the Yerington area [Proffett, 1977; Geissman et al., 1982a, b] to the northwest. Our sampling localities cover different structural subregions of west-central Nevada (e.g., the Gillis Canyon pluton, centrally located in the Walker Lane; the Water Canyon and Gunmetal plutons, within a simple horst block at the southernmost end of the Walker Lane), and we infer that the distribution of the data in Figure 10a in part reflects irregular, minor post intrusion/remagnetization deformation.

Second, paleomagnetic data have been gathered from some Mesozoic units in

19449194, 1984, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC003i002p00179 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3. Mina Deflection Paleomagnetic Data, "Discordance" Test

Locality	Age, m.y.	Expected Declination, deg	Expected Inclination, deg	A_{95}, deg	R, deg	del R, deg	F, deg	del F, deg
Water Canyon pluton	70	342	65	7	19	10	8	6
Gunmetal pluton	90	332	67	4	22	10	7	6
East Garfield Hills pluton	90	332	67	4	17	33	-7	9
La Panta pluton	90	332	67	4	50	20	1	8
Gillis Canyon pluton	70	342	65	7	12	10	15	6
Excelsior Mountains syncline	100	331	65	10	-22	29	-14	8

The age of the unit(s) investigated at each sampling locality has been rounded to the nearest 10 m.y. The expected directions have been calculated from paleomagnetic pole positions given by Irving and Irving [1982]; A_{95} is the semiangle of the circle of 95% confidence about the reference pole position [Irving and Irving, 1982]. R and F are rotation and flattening values determined from the observed directions (Table 2); del R and del F are their 95% confidence limits [Beck, 1980; Demarest, 1983].

other parts of west-central Nevada. For the present discussion, we note that discrepancies, if present, between expected and observed paleomagnetic data in each of these studies are probably best explained in terms of local tilting, rather than clockwise rotations of a more regional nature. The studies are by Geissman et al. [1982b], on plutons and layered rocks of Jurassic age near Yerington (Figure 1) (data implying two periods of westside downtilting with a cumulative magnitude greater than 100°); N. B. Harris and M. O. McWilliams (manuscript in preparation, 1984) on deformed layered rocks of the Pamlico allochthon and a Cretaceous porphyry dike in the New Boston-Blue Ribbon district south-southwest of Luning (Figure 1) (data implying a Late Cretaceous remagnetization and a simple southeastside downtilt, probably in late Cenozoic time, of about 17° about a northeast tilt axis); and by Kluger-Cohen [1981], on carbonates of the Luning formation in the west-central Shoshone Mountains (Figure 1) (data, after correction for the tilt of the strata, implying little, if any rotation of the units; the age of the magnetization was assumed to be Late Triassic, although no exact field tests were applied).

Finally, if the Mina deflection were related to prolonged dextral shear along a number of northwest trending, en echelon strike-slip faults (Walker Lane) in Late Cretaceous and Tertiary time, it is conceivable that each elongated crustal block within the shear zone would not be rotated in a clockwise fashion. Rather, differential movement to produce the deflection would be taken up as deformation along the shear planes themselves. A model for such an origin is shown in Figures 11a, 11b. Paleomagnetic data precluding significant regional postintrusion clockwise rotation, as reported here, would not provide an unequivocal solution to the nature of a postulated shear zone. We suggest that a model such as that shown in Figure 11b is highly unlikely, however. There is a lack of a sufficient number of throughgoing, northwest trending shears in the immediate Mina area capable of producing such a feature. The pervasive Cenozoic structures parallel the Mina deflection and change orientation from northwesterly to west-southwesterly south of Mina. Also, pre-Tertiary structures and lithofacies boundaries clearly are oriented east-west in the Mina area. The continuous nature of these features could be difficult to preserve in such a model.

The data from layered rocks in the

19449194, 1984, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC003i002p00179 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Mina area have no bearing on the paleolatitude/orientation of pre-Late Cretaceous units forming the Mesozoic marine province of the northern Great Basin [Speed, 1979; Oldow, 1983b; Oldow et al., this issue] and pre-Late Cretaceous structural events. Sedimentary rocks of the Excelsior Mountains syncline yield magnetization data that fail the fold test on a 99.9% confidence level. We suggest that remagnetization took place during postfolding intrusion of the Silver Dyke pluton. Additional data, such as those reported by Russell et al. [1982] from deformed Permian to Cretaceous rocks in the Black Rock Desert (northwest Nevada) and by N. B. Harris and M. O. McWilliams (manuscript in preparation, 1984), imply that Cretaceous remagnetization affected layered units widely throughout the western Great Basin. It will be worthwhile to continue to search for prefolding/remagnetization Mesozoic magnetizations in layered rocks which could be used to infer aspects of earlier regional tectonism in the western Great Basin.

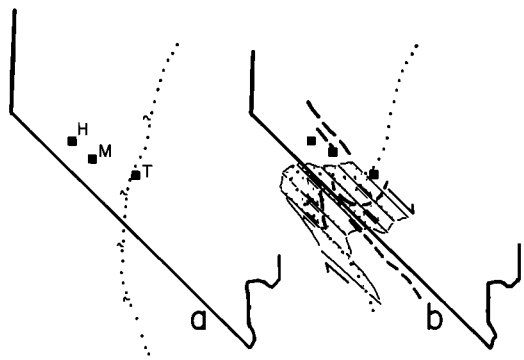


Fig. 11. Model illustrating a potential mechanism for the formation of the Mina deflection region in (a) Late Cretaceous and (b) Tertiary time, relying on deformation by predominantly dextral slip along numerous northwest trending, en echelon faults. In this model, rotational deformation between faults is not required and thus the paleomagnetic declinations of sheared units between faults are not rotated. The dotted line in Figure 11b is the present eastern limit of Permo-Triassic volcanic rocks [Albers, 1967]; its Late Cretaceous position is inferred in Figure 11a.

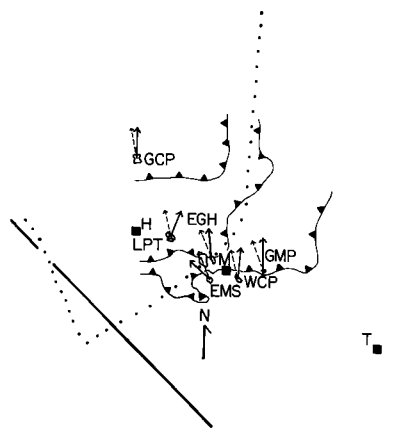


Fig. 12. Paleomagnetic data from basement plutons and remagnetized layered rocks, Mina deflection area, superimposed on a generalized map of Mesozoic structural/tectonic features. Several Jurassic and Cretaceous thrust fault systems are located. Solid arrows refer to the observed mean paleomagnetic declinations found for the respective sampling groups; dashed arrows mark the expected declinations. The arrows are labelled as follows: WCP, Water Canyon pluton; LPT, La Panta pluton; GMP, Gunmetal pluton; EGH, East Garfield Hills pluton; GCP, Gillis Canyon pluton; EMS, Excelsior Mountains syncline. The dotted line is the 0.706 initial $^{87}Sr/^{86}Sr$ isopleth of Kistler [1983] and Kistler et al. [1981].

SUMMARY AND RAMIFICATIONS

The paleomagnetic data from plutons and remagnetized metasedimentary and metavolcanic rocks in the Mina-Hawthorne area of west-central Nevada indicate that only minor, if any, regional clockwise oroflexural bending has occurred in latest Cretaceous and younger time. We summarize our observations in Figure 12, where observed paleomagnetic declinations for each sampling location are plotted with respect to expected directions for respective time periods (Table 3). Specifically, the idea that the Mina deflection formed in association with major Cenozoic tectonic features such as the Walker Lane [Albers, 1967] can probably be ruled out. On a structural basis, we can also suggest that the feature was not caused by post latest Cretaceous reactivation of northwest trending structures with dextral shear. Unfortunately, we cannot

19449194, 1984, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC003i002p00179 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

assess the nature and magnitude of pre-Late Cretaceous deformational events with our paleomagnetic data. However, we find it unlikely that the Mina deflection was formed during earlier Mesozoic events, such as left-lateral slip along the Pine Nut fault system (Figure 1) in Jurassic and Early Cretaceous time [Oldow, 1983b]. The structural work of Oldow [1981a, b, 1984a] on units of the Luning and Roberts Mountains allochthons has shown the consistency of structures related to a primary embayment along the continental margin formed by the Mina deflection. Also, the geochronologic work of Speed and Kistler [1980] has shown that at least some of the contraction in the Mina area was of Early Cretaceous age.

The paleomgnetic data reported here are furthermore consistent with the idea that the Mina deflection existed prior to Middle Devonian and later accretionary tectonic events in the central and western Great Basin. The west stepping bend is probably a primary feature, inherited from the Precambrian breakup of North America. Interpretations of Paleozoic and younger collisional events along the western margin of the North American craton must take into account the irregularities of this margin. For example, Oldow [1984a] has proposed a model for the accretion of the Roberts Mountains allochthon from Middle Devonian to Early Mississippian time which requires that the Mina deflection formed a major embayment along the continental margin. The attempted subduction of the Mina deflection promontory beneath an eastward migrating accretionary arc prism had the effect of reducing the displacement on the basal thrust of the wedge overlying the promontory and the development of cross-folds with axial traces subparallel to the local continental margin. In Mesozoic time, the successor basin west of the Roberts Mountains and Golconda thrust systems [Oldow, 1981a, 1983b, 1984b, Speed, 1978a, 1983] was shortened as layered units of the Luning, Pamlico, and other allochthons were contracted southeastward several tens of kilometers [Oldow, 1981a, b, 1983b; Speed, 1983]. The traces of the thrusts (Figures 1a and 12) were controlled at least in part by the geometry of the Mina deflection [Oldow, 1984b; Oldow et al., this issue]. Concurrent southerly displacement of the Sierran arc and back arc region [Oldow, 1983b, 1984b; Oldow et al., this issue] took place along the left-lateral Pine Nut fault system which probably intersected the western edge of the Mina deflection promontory.

Acknowledgements. We thank M. Geissman, P. Geissman, M. Hudson, J. Kosztowny, and E. Suher for aid in sample collection and preparation. The review comments of M. Beck, Jr., R. Reynolds, R. Schweickert, and an unknown referee are appreciated. The use of the Curie balance at the U.S.G.S. laboratory in Denver is acknowledged. A preprint of the work by Harris and McWilliams is appreciated. Partial support was provided by the Office of Research Development, Colorado School of Mines, by a grant to J. S. O. from the National Science Foundation (grant #EAR-8107673), and by a grant to J. T. C. by Sigma Xi. We thank Union Carbide Co. for K-Ar age data on the Gunmetal pluton.

REFERENCES

Albers, J. P., Belt of sigmoidal bending and right lateral faulting in the western Great Basin, Bull. Geol. Soc. Am., 78, 143-156, 1967.

Beck, M. E., Jr., Paleomagnetic record of plate-margin tectonic processes along the western edge of North America, J. Geophys. Res., 85, 7115-7131, 1980.

Callian, J. T., and J. W. Geissman, Paleomagnetic data from Cretaceous plutons and their bearing on oroflexural deformation in west-central Nevada (abstract), Eos Trans. AGU, 63, 916, 1982.

Carlson, J. E., D. W. Laird, J. A. Peterson, J. H. Schilling, M. L. Silberman, and J. H. Stewart, Preliminary map showing the distribution and isotopic ages of Mesozoic and Cenozoic intrusive rocks in Nevada, U.S. Geol. Surv. Open File Rep., 75-499, 1975.

Demarest, H. H., Jr., Error analysis for the determination of tectonic rotation from paleomagnetic data, J. Geophys. Res., 88, 4321-4328, 1983.

Everndon, J. F., and R. W. Kistler, Chronology of emplacement of Mesozoic batholithic complexes in California and western Nevada, U.S. Geol. Surv. Prof. Pap., 623, 42 pp., 1970.

Farmer, G. L., and D. J. DePaolo, Origin of Mesozoic and Tertiary granite in the western United States and implications for pre-Mesozoic crustal structures, 1, Nd and Sr isotopic studies in the geocline of the northern Great Basin, J.

19449194, 1984, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC003i002p00179 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Geophys. Res., 88, 3379-3402, 1983.
Ferguson, H. G., and S. W. Muller, Structural geology of the Hawthorne and Tonopah Quadrangles, Nevada, U.S. Geol. Surv. Prof. Pap., 216, 55 pp., 1949.
Fisher, R. A., Dispersion on a sphere, Proc. R. Soc. London Ser. A., A217, 295-305, 1953.
Garside, L. J., and M. L. Silberman, New K-Ar ages of volcanic and plutonic rocks from the Camp Douglas Quadrangle, Mineral County, Nevada, Isochron West, 22, 29-32, 1978
Geissman, J. W., R. Van der Voo, W. C. Kelly, and G. H. Brimhall, Jr., Paleomagnetism, rock magnetism, and aspects of structural deformation of the Butte mining district, Butte, Montana, J. Geol., 88, 129-159, 1980.
Geissman, J. W., R. Van der Voo, and K. L. Howard, Jr., A paleomagnetic study of the structural deformation in the Yerington district, Nevada, 1, Tertiary units and their tectonism, Am. J. Sci., 282, 1042-1079, 1982a.
Geissman, J. W., R. Van der Voo, and K. L. Howard, Jr., A paleomagnetic study of the structural deformation in the Yerington district, Nevada, 2, Mesozoic "basement" units and their total and pre-Oligocene tectonism, Am. J. Sci., 282, 1080-1109.
Gillett, S. L., and D. R. Van Alstine, Late Tertiary clockwise oroflexural bending in southern Nevada--Paleomagnetic support of a geologic hypothesis (abstract), Geol. Soc. Am. Abstr. Programs, 14, 166, 1982a.
Gillett, S. L., and D. R. Van Alstine, Remagnetization and tectonic rotation of Upper Precambrian and Lower Paleozoic strata from the Desert Range, southern Nevada, J. Geophys. Res., 87, 10,929-10,954, 1982b.
Gromme, C. S., R. T. Merrill, and J. Verhoogen, Paleomagnetism of Jurassic and Cretaceous plutonic rocks in the Sierra Nevada, California, and its significance for polar wandering and continental drift, J. Geophys. Res., 72, 5661-5684, 1967.
Hardyman, R. F., Volcanic stratigraphy and structural geology of Gillis Canyon Quadrangle, northern Gillis Range, Mineral County, Nevada, Ph.D. dissertation, 268 pp., Univ. of Nev., Reno, 1978.
Hardyman, R. F., Geologic map of the Gillis Canyon Quadrangle, Mineral County, Nevada, Misc. Invest. Map I-1237, U.S. Geol. Surv., Reston, Va.,1980.
Hardyman, R. F., and J. S. Oldow, Structure of allochthonous Mesozoic rocks of the northern Gillis Range, west-central Nevada, Bull. Geol. Soc. Am., in press, 1984.
Irving, E., and G. A. Irving, Apparent polar wander paths Carboniferous through Cenozoic and the assembly of Gondwana, Geophys. Surv., 5, 141-188, 1982.
John, D. A., Map showing the distribution, ages, and petrographic characteristics of Mesozoic plutonic rocks in the Walker Lane 1° by 2° quadrangle, California and Nevada, Misc. Field Stud. Map MF-1382-B, U.S. Geol. Surv., Reston, Va., 1983.
Kanter, L. R., and M. O. McWilliams, Rotation of the southernmost Sierra Nevada, J. Geophys. Res., 87, 3819-3830, 1982.
Kistler, R. W., Configuration of the pre-Middle Jurassic continental margin in the western United States (abstract), Geol. Soc. Am. Abstr. Programs, 15, 272, 1983.
Kistler, R. W., and Z. E. Peterman, Variations in Sr, Rb, K, Na, and initial Sr^{87}/Sr^{86} in Mesozoic granitic rocks and intruded wall rocks in central California, Bull. Geol. Soc. Am., 84, 3489-3512, 1973.
Kistler, R. W., E. D. Ghent, and J. R. O'Neil, Petrogenesis of garnet two-mica granites in the Ruby Mountains, Nevada, J. Geophys. Res., 86, 10,591-10,606, 1981.
Klugher-Cohen, K. K., Utilization of the paleomagnetism of Mesozoic miogeosynclinal and magmatic arc deposition from the North American western cordillera as a test of the sinistral dsplacement along the Mojave-Sonora Megashear, Ph.D. dissertation, 408 pp., Univ. of Pittsburgh, Pittsburgh, Pa., 1981.
Molinari, M. P., Wrench fault tectonics of the southeast margin of the central Walker Lane, west central Nevada (abstract), Geol Soc. Am. Abstr. Programs, 15, 304, 1983.
Muller, S. W., and H. G. Ferguson, Mesozoic stratigraphy of the Hawthorne and Tonopah Quadrangles, Nevada, Bull. Geol. Soc. Am., 50, 1573-1624, 1939.
Nielson, R. L., Geology of the Pilot Mountains and vicinity, Ph.D. dissertation, 161 pp., Univ. of Calif., Berkeley, 1964.

19449194, 1984, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC003i002p00179 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Oldow, J. S., Structure and kinematics of the Pilot Mountains allochthon, W. Nevada, Ph.D. dissertation, 206 pp., Northwestern Univ., Evanston, Ill., 1978a.

Oldow, J. S., Triassic Pamlico Formation: An allochthonous sequence of volcanogenic-carbonate rocks in west-central Nevada, in Mesozoic Paleogeography of the Western United States, edited by D. G. Howell and K. A. McDougall, pp. 223-235, Society of Economic Paleontologists and Mineralogists, Pacific Section, Los Angeles, Calif., 1978b.

Oldow, J. S., Kinematics of late Mesozoic thrusting, Pilot Mountains, west-central Nevada, J. Struct. Geol., 3, 39-49, 1981a.

Oldow, J. S., Structure and stratigraphy of the Luning allochthon and the kinematics of allochthon emplacement, Pilot Mountains, west-central Nevada, Bull. Geol. Soc. Am., 92, 888-911, 1981b.

Oldow, J. S., Late Mesozoic Luning-Fencemaker Thrust system of western Nevada: Regional extent and tectonic implications (abstract), Geol. Soc. Am. Abstr. Programs, 15, 294, 1983a.

Oldow, J. S., Tectonic implications of a late Mesozoic fold and thrust belt in northwestern Nevada, Geology, 11, 542-546, 1983b.

Oldow, J. S., Spatial variability in the structure of the Roberts Mountains allochthon, western Nevada, Bull. Geol. Soc. Am., in press, 1984a.

Oldow, J. S., Evolution of a late Mesozoic back-arc fold and thrust belt, western Great Basin, U.S.A., Tectonophysics, in press, 1984b.

Oldow, J. S., and J. W. Geissman, Oroflexural deformation in west-central Nevada reassessed: Evidence from paleomagnetic data (abstract), Eos Trans. AGU, 63, 309, 1982.

Oldow, J. S., H. G. Ave Lallemant, and W. J. Schmidt, Kinematics of plate convergence deduced from Mesozoic structures in the western Cordillera, Tectonics, this issue.

Proffett, J. M., Jr., Cenozoic geology of the Yerington District, Nevada, and implications for the nature and origin of Basin and Range faulting, Bull. Geol. Soc. Am., 88, 247-266, 1977.

Puffer, J. H., Iron-bearing minerals as indicators of intensive variables pertaining to granitic rocks from the Pegmatite Points area, Colorado, Am. J. Sci., 272, 273-289, 1973.

Reilly, M. B., Petrographic provinces and provenance of the Upper Triassic Luning Formation, west-central Nevada: Summary, Bull. Geol. Soc. Am., 91, 573-575, 1980.

Russell, B. J., M. E. Beck, Jr., R. F. Burmester, and R. C. Speed, Cretaceous magnetizations in northwestern Nevada and tectonic implications, Geology, 10, 423-428, 1982.

Ryall, A., and K. Priestly, Seismicity, secular strain, and maximum magnitude in the Excelsior Mountains area, western Nevada and eastern California, Bull. Geol. Soc. Am., 86, 1585-1592, 1975.

Sanderson, D. D., Spatial distribution and origin of magnetite in an intrusive igneous mass, Bull. Geol. Soc. Am., 85, 1183-1188, 1974.

Schilling, J. H., Isotopic age determinations of Nevada rocks, Nev. Bur. Mines Rep., 10, 79 pp., 1965.

Seidensticker, C. M., Geochemistry and regional correlation of pre-Tertiary volcanic rocks in west-central Nevada, M.Sc. thesis, 96 pp., Rice Univ., Houston, Tex., 1983.

Speed, R. C., Island-arc and other paleogeographic terranes of Late Paleozoic age in the western Great Basin, in Paleozoic Paleogeography of the Western United States, edited by J. H. Stewart, C. H. Stevens, and A. E. Fritche, pp. 349-362, Society of Economic Paleontologists and Mineralogists, Pacific Section, Los Angeles, Calif., 1977.

Speed, R. C., Basinal terrane of the early Mesozoic marine province of the western Great Basin, in Mesozoic Paleogeography of the Western United States, edited by D. G. Howell and K. A. McDougall, pp. 237-252, Society of Economic Paleontologists and Mineralogists, Pacific Section, Los Angeles, Calif., 1978a.

Speed, R. C., Paleogeographic and plate tectonic evolution of the Early Mesozoic marine province of the western Great Basin, in Mesozoic Paleogeography of the Western United States, edited by D. G. Howell and K. A. McDougall, pp. 253-270, Society of Economic Paleontologists and Mineralogists, Pacific Section, Los Angeles, Calif., 1978b

Speed, R. C., Collided Paleozoic microplate in the western United States, J. Geol., 87, 279-292, 1979.

Speed, R. C., Preliminary geologic map of the Sodaville Quadrangle, Mineral County,

19449194, 1984, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC003i002p00179 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Nevada, Misc. Field Stud. Map MF-1300, U.S. Geol. Surv., Reston, Va., 1980.
Speed, R. C., Mesozoic active margin-foreland tectonics in the Great Basin (abstract), Geol. Soc. Am. Abstr. Programs, 15, 271, 1983.
Speed, R. C., and R. W. Kistler, Cretaceous volcanism, Excelsior Mountains, Nevada, Bull. Geol. Soc. Am., 91, 392-398, 1980.
Stewart, J. H., Regional tilt patterns of Late Cenozoic basin-range fault blocks, western United States, Bull. Geol. Soc. Am., 91, 460-464, 1980.
Vetter, U. R., and A. S. Ryall, Systematic change of focal mechanism with depth in the Western Great Basin, J. Geophys. Res., 88, 8237-8250, 1983.
Wetterauer, R. H., Structural analysis of the Jurassic Dunlap Formation, western Nevada (abstract), Geol. Soc. Am. Abstr. Programs, 6, 275, 1974.
Wetterauer, R. H., The Mina deflection--A new interpretation based on the history of the Lower Jurassic Dunlap Formation, western Nevada, Ph.D. dissertation, 159 pp., Northwestern Univ., Evanston, Ill., 1977.
Wu, Y. T., M. Fuller, and V. A. Schmidt, Microanalysis of NRM in a granodiorite intrusion, Earth Planet. Sci. Lett., 25, 275-285, 1974.
Zijderveld, J. D. A., A.C. demagnetization of rocks: Analysis of results, in Methods in Paleomagnetism, edited by D. W. Collinson et al., pp. 254-286, Elsevier, New York, 1967.

J. T. Callian, J. W. Geissman, and S. E. Humphries, Department of Geology, Colorado School of Mines, Golden, CO 80401
J. S. Oldow, Department of Geology, Rice University, Houston, TX 77251

(Received July 15, 1983;
revised November 30, 1983;
accepted December 26, 1983.)

19449194, 1984, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC003i002p00179 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License